Equivalent cyclic-gap, packing and polynomial conditions for equality in the Tu-Deng bound

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Editorial note. Entire original mathematical body, coefficient support proof, narrative equivalence proof and embedded bibliography retained; author owns correctness, no proof repair, added assumptions or independent certification.

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Editorial note. Original cyclic-gap summary is a short restatement of part of the main theorem; all equivalent formulations and the z=1/single-defect cases are one count. Counting corollaries and the cited earlier conjecture-bound proof are not additional targets.

Editorial note. No theorem/proof environments are fabricated for the native narrative outer proof. References to AI assistance in cited authors work remain source context without provenance inference or dependency tracing.

Allowed standard prerequisites (editorial).

- Binary Hamming weights, cyclic gaps and Tu-Deng deletion polynomials. Original S_{t,k}, k-bit convention, zero count, cyclic distance, positive deletion matrices/polynomial and diagonal defect notation retained together with the exact earlier polynomial facts cited by author.. The new gap-support coefficient argument and every equivalence direction are complete in the selected original proof; cyclic-gap summary theorem and counting corollaries are not extra targets.

1 History of the Tu-Deng Conjecture

The Tu-Deng Conjecture stated in the Abstract was first stated in a 2011 paper [9] and soon led to much research in an effort to prove it. The author of the present paper and two of his students at the time soon wrote a paper [3] proving some special cases of the Tu-Deng Conjecture. A little later the present author stated a conjecture which is implied by the Tu-Deng Conjecture (a formal proof of that was given in [6]) and this came to be known as the Cusick Conjecture. The Cusick Conjecture was proved for almost all values of its integer parameter t in [4]. A number of other papers considered the Cusick Conjecture. In [7] it was shown that the Cusick Conjecture is true whenever a certain condition on the cyclic gaps defined in the next section is true. The conjecture was finally proved by Cheng [1], who made use of a random walks idea of Tarlowski [8]. The paper of Cheng almost immediately led to two proofs of the Tu-Deng Conjecture. One was by the present author [2] and the other was by 3 Chinese authors Liu, Luo and Xie [5], who thanked ChatGPT for assisting in the proof.

2 Notation and the meaning of a cyclic gap

Fix $k \geq 2$, put

$$M = 2^k - 1,$$

and let $1 \leq t < M$. As in the Tu-Deng conjecture (see Abstract or [2, Conjecture 1.1]), define

$$S_{t,k} = \left\{(a,b) \in \{0,1,\ldots,M-1\}^2 : a + b \equiv t \pmod{M}, \quad \mathrm{wt}(a) + \mathrm{wt}(b) < k\right\}.$$

Write the k-bit expansion of t as the cyclic word

$$W = \varepsilon_0\varepsilon_1\cdots\varepsilon_{k-1}, \qquad \varepsilon_i \in \{0,1\},$$

and put

$$r = \sum_{i=0}^{k-1} \varepsilon_i = \mathrm{wt}(t), \qquad z = k - r.$$

Since $1 \le t < M$, both r and z are positive.

Let the zero positions, in cyclic order, be

$$q_1, q_2, \ldots, q_z \in \mathbb{Z}/k\mathbb{Z}.$$

Define the cyclic gaps $d_1, \ldots, d_z$ by

$$d_\nu = (q_{\nu+1} - q_\nu) \pmod{k}, \qquad 1 \le d_\nu \le k, \qquad q_{z+1} = q_1.$$

Thus

$$d_1 + \cdots + d_z = k. \tag{1}$$

This convention is important: d_ν is the distance (or gap) from one zero to the next zero, so it is one more than the number of intervening ones. Thus, if there are a ones between two consecutive zeros, the corresponding gap is $a+1$. For example, the two gaps of the cyclic word 0101 are both 2.

Set

$$g(W) = \min_{1\le\nu\le z} d_\nu. \tag{2}$$

3 The main equivalence theorem

The simplest necessary and sufficient condition for $|S_{t,k}| = 2^{k-1}$ can be stated very briefly in terms of the cyclic gaps as follows:

Theorem 3.1 (Cyclic gap characterization)**.** *Let $1 \le t < 2^k - 1$, and let $z = k - \mathrm{wt}(t)$. In the cyclic k-bit expansion of t, let $d_1, \ldots, d_z$ be the distances between successive zeros, where the distance includes the final zero. Then*

$$|S_{t,k}| = 2^{k-1} \quad\Longleftrightarrow\quad \min_\nu d_\nu \ge z.$$

Equivalently, at least $z-1$ ones occur between every two consecutive zeros.

The next theorem will give a long list of necessary and sufficient conditions. Some are purely combinatorial, while the last three use the polynomial notation of the proof in [2, Sections 2–5 and 8].

With the matrices from [2, Sections 2–3],

$$C_0 = \begin{pmatrix} 1 & 0 \\ u & v \end{pmatrix}, \qquad C_1 = \begin{pmatrix} u & v \\ 0 & 1 \end{pmatrix}, \qquad M_W = C_{\varepsilon_0} C_{\varepsilon_1} \cdots C_{\varepsilon_{k-1}},$$

put

$$Q_W(u,v) = \mathrm{tr}\, M_W - u^r v^z.$$

Let

$$H_W(u,v) = \sum_{i,j\ge 0} h_{i,j}(W) u^i v^j$$

be the polynomial of [2, equation (10)], defined by

$$Q_W(u,v) = 1 + (u+v-1)H_W(u,v), \tag{3}$$

and define the positive deletion polynomial

$$R_W(u,v) = H_W + D_u H_W, \qquad D_u = (u-1)\frac{\partial}{\partial u} + v\frac{\partial}{\partial v}. \tag{4}$$

Also define the diagonal defects

$$\Delta_m^-(W) = 2h_{m-1,m}(W) - h_{m,m}(W), \qquad m \ge 1. \tag{5}$$

Theorem 3.2 (Equivalent equality conditions)**.** *The following conditions are equivalent.*

(i) $|S_{t,k}| = 2^{k-1}$.

(ii) *Every cyclic gap satisfies*
$$d_\nu \geq z.$$

(iii) *Every zero is isolated and every maximal cyclic run of ones has length at least* $z-1$.

(iv) *The cyclic word* W *contains no cyclic factor*
$$0\,1^j\,0 \qquad (0 \leq j \leq z-2).$$
When $z = 1$*, the displayed range of* j *is empty.*

(v) *If*
$$Z = \{i \in \mathbb{Z}/k\mathbb{Z} : \varepsilon_i = 0\},$$
then
$$Z \cap (Z+s) = \varnothing \qquad (1 \leq s \leq z-1). \tag{6}$$
Equivalently, the z *cyclic intervals*
$$q + [0, z-1] \pmod{k}, \qquad q \in Z,$$
are pairwise disjoint.

(vi) *There are nonnegative integers* $e_1, \ldots, e_z$ *such that*
$$d_\nu = z + e_\nu \quad (1 \leq \nu \leq z), \qquad e_1 + \cdots + e_z = k - z^2. \tag{7}$$

(vii) *The positive deletion polynomial satisfies*
$$u^{z-1} \mid R_W(u, v). \tag{8}$$

(viii) *The Laurent polynomial* $R_W(x, x^{-1})$ *has no negative powers of* x.

(ix) *Every diagonal defect vanishes:*
$$\Delta_m^-(W) = 0 \quad (m \geq 1),$$
or, equivalently,
$$2h_{m-1,m}(W) = h_{m,m}(W) \quad (m \geq 1). \tag{9}$$

If $z \geq 2$*, these conditions are also equivalent to the following single defect test:*
$$r \geq z \quad \textit{and} \quad 2h_{z-2,z-1}(W) = h_{z-1,z-1}(W). \tag{10}$$

For $z = 1$*, equality always holds.*

Thus the proposed condition is exactly condition (ii). In words, if $z = k - \mathrm{wt}(t)$, equality holds precisely when at least $z-1$ ones occur between every two consecutive zeros.

4 Polynomial facts used in the proof

For a nonzero polynomial $P(u, v)$, let $\mathrm{ord}_u P$ denote the least exponent of u occurring in P, and put $\mathrm{ord}_u 0 = \infty$. We record the coefficient and support facts needed below. The point of the lemma is that a short cyclic gap produces a strictly positive coefficient on the first diagonal below the main diagonal.

Lemma 4.1 (Gap-support lemma)**.** *Let* $g = g(W)$.

(a) *The* v*-degree of* R_W *is at most* $z-1$.

(b) *If* $z \geq 2$*, then*
$$\mathrm{ord}_u R_W = g - 1. \tag{11}$$
In particular,
$$u^{z-1} \mid R_W \quad \Longleftrightarrow \quad g \geq z. \tag{12}$$

(c) *Suppose $z \geq 2$, $r \geq z$, and $g < z$. Then*

$$[u^{z-2}v^{z-1}]R_W > 0. \tag{13}$$

Equivalently,

$$\Delta^{-}_{z-1}(W) > 0. \tag{14}$$

Proof. The bit-complement symmetry proved in [2, proof of Theorem 5.2] interchanges u and v. Applying this symmetry to the top-boundary formula in [2, Theorem 8.3, equation (43)] gives

$$\deg_v H_W \leq z - 1.$$

None of the operations in (4) increases the v-degree. Hence

$$\deg_v R_W \leq z - 1, \tag{15}$$

which proves part (a).

We next explain the u-order. The positive differential decomposition in [2, Theorem 5.1, equation (27)] is

$$R_W = \sum_{\varepsilon_i=1} \gamma_{V(i)} + \sum_{\varepsilon_i=0} \delta_{V(i)}, \tag{16}$$

where $V(i)$, γ_V, and δ_V are those of [2, Sections 2 and 5]. All the summands have nonnegative coefficients. The relevant part of the six-state recurrence from [2, Theorem 2.2, equations (6)–(7)] is

	γ'	δ'	d'
append 1	$u\gamma$	$\delta + v\gamma$	ud
append 0	$\gamma + d + u\delta$	$v\delta$	$vd.$

(17)

Put

$$A = \operatorname{ord}_u \gamma, \qquad B = \operatorname{ord}_u \delta, \qquad C = \operatorname{ord}_u d,$$

with $\operatorname{ord}_u 0 = \infty$. Taking u-orders in (17) gives

	A'	B'	C'
append 1	$A + 1$	$\min\{B, A\}$	$C + 1$
append 0	$\min\{A, C, B + 1\}$	B	$C.$

(18)

First suppose that the zeros of W are isolated. After a cyclic rotation write

$$W = 0\,1^{a_1}0\,1^{a_2}\cdots 0\,1^{a_z}, \qquad a_\nu \geq 1. \tag{19}$$

Applying (18) successively to the blocks in (19) gives the following two elementary facts:

$$\begin{aligned} &\operatorname{ord}_u \delta_{V(i)} = a_\nu, \quad \text{if the deleted position } i \text{ is the zero preceding the run } 1^{a_\nu}, \\ &\operatorname{ord}_u \gamma_{V(i)} = a_\nu, \quad \text{if the deleted position } i \text{ lies in the run } 1^{a_\nu}. \end{aligned} \tag{20}$$

Here is the short induction behind (20). Begin with $(A, B, C) = (\infty, \infty, 0)$. If the deleted position is the zero preceding 1^{a_ν}, then $V(i)$ begins with a_ν ones. After these ones and the next zero, (18) gives $A = a_\nu$ and $B = \infty$; at the first following one, B becomes a_ν. Thereafter, before every later zero one has $A, C \geq a_\nu + 1$, so that the zero resets A to $a_\nu + 1$ while leaving B unchanged. Thus a later one can never lower B, and the final value is $B = a_\nu$.

If instead the deleted position is a one in 1^{a_ν}, write p for the number of ones of that run occurring at the beginning of $V(i)$ and q for the number occurring at the end. Then $p + q = a_\nu - 1$. After the initial p ones and the next zero, (18) gives $A = p$, $B = \infty$, and $C = p$. At the first subsequent one, B becomes p and remains p. At each later zero, after the first, the value of A is reset to $p + 1$: immediately before such a zero one has $A \geq p + 1$ and $C \geq p + 1$, while $B + 1 = p + 1$. In particular, at the zero preceding the terminal block of q ones we have $A = p + 1$. The final q ones increase A to $p + 1 + q = a_\nu$. This proves both assertions in (20).

Since (16) is a sum of polynomials with nonnegative coefficients, (20) implies

$$\operatorname{ord}_u R_W = \min_\nu a_\nu = g - 1. \tag{21}$$

If two zeros are adjacent, then $g = 1$. The calculation below shows that $[u^0 v^{z-1}]R_W > 0$, and therefore $\operatorname{ord}_u R_W = 0 = g - 1$ in that case also. This proves part (b).

It remains to prove the asserted strict coefficient. Write the general cyclic run decomposition as

$$W = 1^{a_1}0^{b_1}\cdots 1^{a_p}0^{b_p}, \qquad a_\nu, b_\nu \geq 1, \tag{22}$$

where

$$a_1 + \cdots + a_p = r, \qquad b_1 + \cdots + b_p = z.$$

Define

$$P(y) = \prod_{\nu=1}^{p}(1 + y + \cdots + y^{a_\nu}) = \sum_{i=0}^{r} A_i y^i. \tag{23}$$

Using again the bit-complement symmetry from [2, proof of Theorem 5.2] and then the top-boundary formula [2, Theorem 8.3, equation (43)] gives

$$h_{i,z-1}(W) = A_i \quad (0 \leq i < r), \qquad h_{r,z-1}(W) = 0. \tag{24}$$

From (4),

$$[u^i v^j]R_W = (i + j + 1)h_{i,j} - (i + 1)h_{i+1,j}. \tag{25}$$

Consequently, for $0 \leq i < r$,

$$[u^i v^{z-1}]R_W = (i + z)A_i - (i + 1)A_{i+1}. \tag{26}$$

Put $F_a(y) = 1 + y + \cdots + y^a$. The elementary identity

$$F_a - (1 - y)F_a' = (a + 1)y^a \tag{27}$$

implies

$$pP - (1 - y)P' = \sum_{\nu=1}^{p}(a_\nu + 1)y^{a_\nu} \prod_{\substack{\mu=1 \\ \mu\neq\nu}}^{p} F_{a_\mu}(y). \tag{28}$$

Therefore

$$\begin{aligned} zP - (1 - y)P' = (z - p)P \\ + \sum_{\nu=1}^{p}(a_\nu + 1)y^{a_\nu} \prod_{\substack{\mu=1 \\ \mu\neq\nu}}^{p} F_{a_\mu}(y). \end{aligned} \tag{29}$$

The coefficient of y^i on the left side of (29) is exactly the right side of (26).

If some two zeros are adjacent, then $p < z$. Since every coefficient of P from degree 0 through degree r is positive, the term $(z - p)P$ shows that

$$[u^i v^{z-1}]R_W > 0 \qquad (0 \leq i < r). \tag{30}$$

In particular, taking $i = 0$ proves the adjacent-zero assertion used in the proof of part (b).

Suppose instead that all zeros are isolated. Then $p = z$, and the one-run lengths in (22) are precisely the integers $d_\nu - 1$. If $g < z$, some $a_\nu = g - 1 \leq z - 2$. Take $i = z - 2$ in (29). Since $r \geq z$, we have $i < r$. Moreover $0 \leq i - a_\nu \leq r - a_\nu$, and a product of polynomials $1 + y + \cdots + y^{a_\mu}$ has a positive coefficient in every degree from 0 through the sum of its degrees. Hence the coefficient of

$$y^{i-a_\nu} \quad \text{in} \quad \prod_{\mu\neq\nu} F_{a_\mu}(y)$$

is positive. Thus (26) again gives

$$[u^{z-2}v^{z-1}]R_W > 0.$$

Finally, putting $m = z - 1$ in (25) gives

$$[u^{z-2}v^{z-1}]R_W = (z - 1)\big(2h_{z-2,z-1} - h_{z-1,z-1}\big) = (z - 1)\Delta_{z-1}^{-}, \tag{31}$$

which proves parts (c) and (14). □

5 Proof of the equivalence theorem

The equivalence among the elementary combinatorial conditions is immediate, but we give the details. Since $d_\nu - 1$ is the number of ones between consecutive zeros,

$$d_\nu \geq z \quad \Longleftrightarrow \quad d_\nu - 1 \geq z - 1. \tag{32}$$

This proves the equivalence of conditions (ii), (iii), and (iv). Condition (v) says exactly that no two zero positions are at cyclic distance $1, 2, \ldots, z-1$, so it is equivalent to condition (ii). Finally, using (1), condition (ii) is equivalent to writing

$$d_\nu = z + e_\nu, \qquad e_\nu \geq 0,$$

where

$$\sum_{\nu=1}^{z} e_\nu = \sum_{\nu=1}^{z} d_\nu - z^2 = k - z^2.$$

Thus conditions (ii)–(vi) are equivalent.

By Lemma 4.1(b), condition (ii) is equivalent to condition (vii) when $z \geq 2$. When $z = 1$, both conditions are automatic, so the equivalence holds in that case as well.

We next connect these statements to equality in the Tu-Deng bound. The coefficient calculation in [2, proof of Theorem 5.2] gives

$$[u^{m-1}v^m]R_W = m\Delta_m^-(W). \tag{33}$$

As in [2, equation (39)], let $\sigma^+(W)$ and $\sigma^-(W)$ denote the total coefficients of the nonnegative and negative powers, respectively, of $Q_W(x/2, x^{-1}/2)$. The exact median identity in [2, Lemma 8.2, equation (41)] is

$$\sigma^+(W) = \frac{1}{2} + \sum_{m \geq 1} 2^{-2m-1}\Delta_m^-(W), \tag{34}$$

and every $\Delta_m^-(W)$ is nonnegative. Hence

$$\sigma^+(W) = \frac{1}{2} \quad \Longleftrightarrow \quad \Delta_m^-(W) = 0 \text{ for every } m \geq 1. \tag{35}$$

Suppose first that the gap condition holds. From (1),

$$k = \sum_{\nu=1}^{z} d_\nu \geq z^2. \tag{36}$$

If $z \geq 2$, this gives

$$r = k - z \geq z(z-1) \geq z. \tag{37}$$

The case $z = 1$ also has $r \geq z$. Lemma 4.1 gives

$$u^{z-1} \mid R_W, \qquad \deg_v R_W \leq z - 1. \tag{38}$$

Consequently R_W has no monomial $u^{m-1}v^m$: if $m \leq z-1$, its u-degree is too small, while if $m \geq z$, its v-degree is too large. Equations (33)–(35) therefore give

$$\sigma^+(W) = \sigma^-(W) = \frac{1}{2}. \tag{39}$$

Theorem 6.1, the final assertion of Theorem 6.2 (following equation (35)), and equation (38) of [2] combine to give

$$\frac{|S_{t,k}|}{2^k} = \mathrm{Neg}\left(Q_W(x/2, x^{-1}/2) + 2^{-k}x^{2r-k}\right), \tag{40}$$

where Neg denotes the total coefficient of the negative powers of x. Since

$$2r - k = r - z \geq 0, \tag{41}$$

the last monomial in (40) contributes no negative mass. Equations (39) and (40) give

$$|S_{t,k}| = 2^{k-1}. \tag{42}$$

Conversely, suppose that the gap condition fails. If $r < z$, then $2r < k$. Equation (44), following Theorem 8.3 in [2], gives

$$\Delta_r^-(W) \geq 2. \tag{43}$$

It follows from (34) that the negative mass of $Q_W(x/2, x^{-1}/2)$ is at most

$$\frac{1}{2} - 2^{-2r}.$$

The determinant monomial in (40) now has negative exponent and adds 2^{-k}. Since $k > 2r$,

$$2^{-k} < 2^{-2r},$$

so the total negative mass in (40) is strictly smaller than $1/2$. Therefore

$$|S_{t,k}| < 2^{k-1}. \tag{44}$$

It remains to consider a failed gap condition with $r \geq z$. By Lemma 4.1(c),

$$\Delta_{z-1}^-(W) > 0.$$

Thus (34) is strict, so $\sigma^+(W) > 1/2$ and $\sigma^-(W) < 1/2$. The determinant monomial has exponent $r - z \geq 0$ and adds no negative mass. Equation (40) again gives the strict inequality (44). This proves

$$|S_{t,k}| = 2^{k-1} \quad \Longleftrightarrow \quad d_\nu \geq z \text{ for every } \nu. \tag{45}$$

The same argument proves the remaining polynomial equivalences. If the gap condition holds, (38) shows that $R_W(x, x^{-1})$ has no negative power. If the gap condition fails and $r < z$, (43) and (33) produce the term $u^{r-1}v^r$; if $r \geq z$, Lemma 4.1(c) produces the term $u^{z-2}v^{z-1}$. Thus a failed gap condition always produces a negative power of x. This proves the equivalence with condition (viii). Equation (33), together with the already established equivalence between the gap condition and equality and with (34)–(35), proves the equivalence with condition (ix).

Finally, suppose $z \geq 2$. If equality holds, we have $r \geq z$ and all defects vanish, so (10) holds. Conversely, suppose (10) holds. If the gap condition failed, Lemma 4.1(c) would give $\Delta_{z-1}^- > 0$, contrary to (10). Thus (10) is equivalent to all of the other conditions. The proof of Theorem 3.2 is complete.

6 Consequences and examples

Corollary 6.1 (A simple numerical obstruction)**.** *If $|S_{t,k}| = 2^{k-1}$ and the k-bit word for t has z zeros, then*

$$z^2 \leq k. \tag{46}$$

Equivalently, an equality word can have at most $\lfloor \sqrt{k} \rfloor$ zeros.

Proof. This is (36). □

If $k = z^2$, all the nonnegative integers e_ν in (7) must be zero. Thus

$$d_1 = \cdots = d_z = z, \tag{47}$$

and the equality words are precisely the rotations of

$$(0\,1^{z-1})^z. \tag{48}$$

There is also a simple enumeration formula.

Corollary 6.2 (Number of equality words with z zeros)**.** *Let $N(k, z)$ be the number of k-bit words on the labeled cyclic positions $\mathbb{Z}/k\mathbb{Z}$ with exactly z zeros for which $|S_{t,k}| = 2^{k-1}$. Then*

$$N(k,z) = \begin{cases} \dfrac{k}{k - z(z-1)} \dbinom{k - z(z-1)}{z}, & k \geq z^2, \\ 0, & k < z^2. \end{cases} \tag{49}$$

Proof. Assume $k \geq z^2$ and put $L = k - z^2$. By condition (vi), after distinguishing one zero, an equality word is specified by a weak composition

$$L = e_1 + \cdots + e_z.$$

There are

$$\binom{L+z-1}{z-1}$$

such compositions and k choices for the position of the distinguished zero. Every word is counted once for each of its z zeros. Hence

$$N(k,z) = \frac{k}{z}\binom{k-z^2+z-1}{z-1}.$$

Using

$$\frac{1}{z}\binom{n-1}{z-1} = \frac{1}{n}\binom{n}{z}, \qquad n = k - z(z-1),$$

gives (49). □

k	cyclic word W	cyclic gaps	conclusion
4	0101	$(2,2)$	$\lvert S_{t,4}\rvert = 8$
5	01011	$(2,3)$	$\lvert S_{t,5}\rvert = 16$
5	00111	$(1,4)$	$\lvert S_{t,5}\rvert = 12 < 16$
9	011011011	$(3,3,3)$	$\lvert S_{t,9}\rvert = 256$

In the first example, the number of intervening ones in each gap is 1, but the cyclic gap is 2. This illustrates why the precise convention for the word "gap" matters.

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